	ground
2(a)(i)	*Radian* describes the plane angle subtended by a circular arc as the length of the arc divided by the radius of the arc. One *radian* is the angle subtended at the center of a circle by an arc that is equal in length to the radius of the circle. Hence, a full circle is 2π radians.
2(a)(ii)	Angular frequency refers to number of radians per second in Simple Harmonic Motion. Each complete oscillation is 2π radians. Hence, the angular frequency gives the number complete oscillations per unit time.
2(b)(i)	Since the motion is SHM, total energy of sphere is constant. $TE = KE_{\max} = PE_{\max}$ $PE_{\max} = mgh = 0.12 \times 9.81 \times 0.004 = 4.7 \times 10^{-3}\,\text{J}$

2(b)(ii)	$KE_{max} = PE_{max}$ $\frac{1}{2} \times m \times v_{max}^2 = 4.7 \times 10^{-3}$ $v_{max}^2 = \frac{2 \times 4.7 \times 10^{-3}}{0.12} = \omega^2 (x_0^2 - x^2)$ $\omega^2 (0.08^2 - 0) = \frac{2 \times 4.7 \times 10^{-3}}{0.12}$ $(2\pi f)^2 = \frac{2 \times 4.7 \times 10^{-3}}{0.12 \times 0.08^2}$ $f = \sqrt{\frac{2 \times 4.7 \times 10^{-3}}{0.12 \times 0.08^2 \times (2\pi)^2}} = 0.56 \text{ Hz}$
3(a)	Change in the internal energy of the system is equal to the summation of the